Week 5 Assignment 5

The due date for submitting this assignment has passed.

Due on 2025-08-27, 23:59 IST.

Assignment submitted on 2025-08-27, 16:00 IST

1) Consider the graph with *V={1,2,3};E={(1,2),(2,1),(1,3),(3,1),(2,3)}* with weights *{2.8,9,1,6}* respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following: ***1 point***

The first row of A_0 is {__,__,__}

- {0,2,9}
- {2,0,9}
- {0,9,2}
- {9,2,0}

Yes, the answer is correct.
Score: 1

Accepted Answers:
{0,2,9}

2) Consider the graph with *V={1,2,3}; E={(1,2),(2,1),(1,3),(3,1),(2,3)}* with weights *{2.8,9,1,6}* respectively. Solve APSP problem on *V* by using Floyd-Warshall algorithm and answer following: ***1 point***

The second row of A_0 is {__,__,__}

- {6,0,8}
- {8,0,6}
- {8,6,0}
- {0,8,6}

Yes, the answer is correct.

Score: 1
Accepted Answers:
{8,0,6}

3) Consider the graph with *V={1,2,3}; E={(1,2),(2,1),(1,3),(3,1),(2,3)}* with weights *{2.8,9,1,6}* respectively. Solve APSP problem on *V* by using Floyd-Warshall algorithm and answer following: **1 po**

The first row of A_1 is {__,__,__}

- {2,0,9}
- {0,9,2}
- {9,2,0}
- {0,2,9}

Yes, the answer is correct.
Score: 1
Accepted Answers:
{0,2,9}

4) Consider the graph with V={1,2,3}; E={(1,2),(2,1),(1,3),(3,1),(2,3)} with weights {2.8,9,1,6} respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following: The third row of A1 is {__,__,__} ***1 point***

- {1,0,3}
- {0,3,1}
- {1,3,0}
- {3,1,0}

Yes, the answer is correct.
Score: 1
Accepted Answers:
{1,3,0}

5) Consider the graph with *V={1,2,3}; E={(1,2),(2,1),(1,3),(3,1),(2,3)}* with weights *{2.8,9,1,6}* respectively. Solve APSP problem on *V* by using Floyd-Warshall algorithm and answer following: ***1 point***

The first row of A_2 is {__,__,__}

- {2,0,8}
- {0,2,8}
- {8,2,0}
- {0,8,9}

Yes, the answer is correct.
Score: 1
Accepted Answers:
{0,2,8}

6) Consider the graph with *V={1,2,3}; E={(1,2),(2,1),(1,3),(3,1),(2,3)}* with weights *{2.8,9,1,6}* respectively. Solve APSP problem on *V* by using Floyd-Warshall algorithm and answer following: ***1 point***

The second row of A_3 is {__,__,__}

- {7,0,6}

Vertex ()

Video download ()

Live session ()

- {0,7,6}
- {7,2,0}
- {0,7,9}

Yes, the answer is correct.
Score: 1
Accepted Answers:
{7,0,6}

7) The complexity of Floyd-Warshall Algorithm for a graph G with b vertices and n edges is θ(__) ***1 point***

- $n^{1/3}$
- $n^{1/2}$
- n^3
- n^2

Yes, the answer is correct.
Score: 1
Accepted Answers:
n^3

8) In converting the APSP problem to a matrix multiplication problem, we use special interpretations for matrix operation. ***1 point***
The operation '+' is interpreted as ____

- Minimum
- Maximum
- Neither minimum nor maximum
- Both minimum and maximum

Yes, the answer is correct.
Score: 1
Accepted Answers:
Minimum

9) In converting the APSP problem to a matrix multiplication problem, we use special interpretations for matrix operation. ***1 point***
The operation '.' is interpreted as ____

- Addition
- Multiplication
- Dot product
- None of the options given here

Yes, the answer is correct.
Score: 1
Accepted Answers:
Addition

10) The complexity of Johnson’s Algorithm for APSP is θ(__) ***1 point***

- ( ) $nlogn + nm$
- (•) $n^2 logn + nm$
- ( ) $n^2 logn - nm$
- ( ) $n^2 lognm + nm$

Yes, the answer is correct.
Score: 1

Accepted Answers:
$n^2 logn + nm$